

Erlang loss models for the static deployment of ambulances

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Abstract How should one allocate a fleet of ambulances to fixed bases with the goal of keeping response times to calls as small as possible? We present two new models for this problem, both of which are based on the Erlang loss formula. The first model is stylized, and shows that allocating ambulances in proportion to the offered load is not necessarily optimal and can often be substantially improved upon. The second model can be used to screen potential allocations to try to identify top candidates for further investigation. Computational experiments on the first model provide insights on how we should modify ambulance allocations in response to different levels of offered load. Computational experiments on the second model compare this model with the so-called A-hypercube model and show that our model has comparable, and in many cases, better performance in terms of the accuracy of the estimates of performance measures. Thus, our models can be used as pre-screening tools to identify promising ambulance allocations and these promising ambulance allocations can subsequently be evaluated carefully through simulation.

Keywords Ambulance allocation · Static deployment · EMS management · Erlang loss · Ambulance base · Lost calls · Performance measures · Spatial queues

1 Introduction

Emergency medical service (EMS) providers face the problem of allocating a fixed number of ambulances among a set of bases. The ultimate goal is to ensure the best possible medical outcomes for patients. Though this is difficult to measure, an easily-measured proxy is the response time for a call, defined as the elapsed time from when a call is received to when an ambulance arrives at the scene. An industry standard is to attempt to ensure that a percentage x of all calls have response times under y minutes, with common choices being on the order of $x = 80$ and $y = 10$. In this paper we provide two new computational models that can help with the ambulance allocation problem, where the goal is to minimize the fraction of calls with response times over a time threshold (calls wRTOTS). We consider only *static* deployments, in the sense that ambulances return to their assigned bases whenever they are available to serve calls. This is in contrast to *dynamic* deployments, where ambulances are directed to different locations throughout their shifts in an attempt to better match anticipated demand.

Our first model, which we call the *island* model, is *prescriptive* in nature, meaning that it implicitly searches over all possible allocations of ambulances to bases, and returns the one that yields the best performance. This ability to search over all possible allocations comes at a price, however. The island model uses a simplified objective function which, while related to

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the goal of minimizing the fraction of calls wRTOTS, is not an exact match for it. It also assumes that each base operates independently of all other bases, so that each base is an “island.” The island model should therefore be viewed as a tool for obtaining insight into effective allocations, but its predictions should be refined through the use of more detailed methods.

Our second model, which we call the *overflow* model, can provide such refined predictions for a proposed allocation of ambulances to bases. As such, it is *descriptive* rather than *prescriptive* in that it provides performance predictions for a single proposed allocation. The overflow model runs quickly (less than a second), and so it can be viewed as a tool for quickly screening out poor ambulance allocations and identifying promising allocations for further investigation. Both the island and overflow models are based on the Erlang loss formula, which is a standard result from queueing theory that has been used many times in EMS modeling, but not in the way we propose here.

Static ambulance deployment problems have received a great deal of attention in the literature. We present a brief survey here to give a feel for the primary approaches. For more detailed surveys, we recommend the excellent reviews by Swersey [27], Brotcorne et al. [4], Goldberg [10] and Green and Kolesar [13].

As mentioned above, prescriptive methods search over all possible ambulance allocations to identify promising allocations, and they typically use simplified performance measures to allow for efficient search procedures. Important early work on prescriptive models includes Toregas et al. [28] and Church and ReVelle [6]. The later model by Daskin [7] attempts to capture some of the stochastic aspects of the problem under the assumption that the ambulances are statistically independent. Batta et al. [1] build on this model to relax the independence assumption, whereas ReVelle and Hogan [24] and Marianov and ReVelle [23] extend the earlier models by adding constraints on the minimum number of ambulances required in various zones. The last two papers are related to our work in the sense that the minimum number of ambulances are obtained from the Erlang loss formula. More recently, Erdogan et al. [8] incorporate medical outcomes into the objective function by concentrating on the lengths of the response times rather than the fraction of calls with response times below a certain time standard. All of the papers mentioned in this paragraph thus far generally use linear integer programming formulations, but nonlinear models have also been used by several authors. For example, Goldberg and Paz [11] embed a heuristic search procedure into the approximate hypercube model proposed by Larson [22]. The underlying

theme of this work is to use tractable approximations for the performance of the system to guide a search procedure.

Descriptive methods provide more accurate performance predictions than are possible with the simplified objective functions used in prescriptive models. This area is dominated by the hypercube method [21] and its extensions, which include Larson [22], Jarvis [18, 19], Berman and Larson [2], Brandeau and Larson [3], Goldberg and Szidarovsky [12] and Budge et al. [5]. Another recent work, also based on queueing theory, is Singer and Donoso [26]. Finally, another important descriptive approach is, of course, simulation. Simulation has been used in a large portion of ambulance deployment studies either to analyze the performance of other models or as a tool in and of itself. For an overview, see Henderson and Mason [16].

The solution to the two-base version of the island model provides useful insights. In particular, it turns out that it is not optimal to allocate ambulances to bases in proportion to the call loads offered to bases. Instead, bases with lighter call loads should receive more than a proportional share of ambulances. Why is this important? This is essentially the problem faced by a central planner who must decide how many vehicles to allocate between two (or more) cities, where the two cities are far enough apart that vehicles based in one city cannot routinely attend calls in the other city. One might be tempted to allocate vehicles in proportion to the load, but our results show that this can be far from optimal. A similar situation arises in cities that, due to traffic congestion and/or geographic layout can be broken down into 2 or more regions between which it is hard to travel. This is precisely the case for the city of Auckland in New Zealand for example, where a major bridge linking the north and south portions of the city is virtually impassable during peak traffic periods. As the peak traffic periods approach, dispatchers might try to dispatch vehicles so as to appropriately distribute ambulances between the two regions.

This island model adds insight, but the assumption that bases do not interact is unfortunate if one wishes to apply the results to a typical city where interaction between bases is substantial. The overflow model relaxes this assumption and is, therefore, less tractable. The overflow model approximates, for a given ambulance allocation, the probability that a given base responds to calls originating in a given district of the city, for all base-district pairs. These probabilities can be used to compute the fraction of calls wRTOTS. This model can also be used to estimate other performance measures, as discussed later. The main idea is to use the Erlang loss formula with carefully chosen parameters

to approximate the probability that all ambulances at one or more bases are busy. These probabilities are then used to compute the probability that a call will be answered by an ambulance at a particular base. The approach is essentially a fixed point scheme for a nonlinear system of equations and it is similar in spirit to the work of Larson [22], Jarvis [18], Goldberg and Szidarovsky [12] and Budge et al. [5] on the approximate hypercube model and its extensions. An important distinction of our work is that the earlier work formulates fixed point equations that involve a sophisticated version of the so-called Q factors, whereas our model avoids the need for Q factors by using the Erlang loss function. A similar use of the Erlang loss function to estimate overflow rates in multi-skill call centers appears in Koole and Talim [20]. Although we do not provide a proof of convergence of the resulting iterative scheme, the computational results suggest that there exists a unique fixed point and the fixed point can be computed extremely quickly. Therefore, as noted above, the overflow model can be used to pre-screen a large number of ambulance allocations, after which the best allocations can be studied through simulation.

This paper is organized as follows. Section 2 introduces the island model, and we study the two-base version of the island model in detail in Section 3. Section 4 describes the overflow model, and Section 5 compares performance estimates obtained via the overflow model, the approximate Hypercube model and a simulation model. We conclude in Section 6.

2 The island model

In this section we present a prescriptive model (the island model) for the static ambulance deployment problem. We have N ambulances to allocate among a set of bases. The set of bases is $\mathcal{B}$ and we can allocate at most c_b ambulances to base b . An ambulance deployed at base b can serve calls at rate μ_b and the calls that are offered to base b arrive according to a Poisson process with rate λ_b .

Our goal here is to obtain *insight* about the problem rather than to obtain predictions that could be used immediately in real organizations. Accordingly, we adopt a model that is mathematically tractable at the expense of some realism. In the island model, each base operates independently. Calls that arrive for a base when all ambulances that are located there are busy are lost, meaning that they are assumed handled by some outside agency (e.g., the fire department, or

a competing EMS organization). So in the island model bases do not handle the overflow calls from other bases, and $\{\lambda_b : b \in \mathcal{B}\}$ are known parameters that do not depend on the ambulance allocation. We relax this strong assumption later in Section 4.

If we allocate n_b ambulances to base b , then each base operates as an $M/G/n_b/n_b$ queue. This implies that we can compute the probability that an arriving call offered to base b finds all n_b ambulances busy by using the Erlang loss formula

$$\mathcal{E}(n_b, \lambda_b/\mu_b) = \frac{[\lambda_b/\mu_b]^{n_b}/n_b!}{\sum_{j=0}^{n_b} [\lambda_b/\mu_b]^j/j!}.$$

See, for instance, Gross and Harris [14] for proof that this formula is valid for general service time distributions.

In this case, we can solve the problem

$$\min \sum_{b \in \mathcal{B}} \lambda_b \mathcal{E}(n_b, \lambda_b/\mu_b) \quad (1)$$

$$\text{subject to } \sum_{b \in \mathcal{B}} n_b = N \quad (2)$$

$$0 \leq n_b \leq c_b \quad b \in \mathcal{B} \quad (3)$$

$$n_b \text{ integer} \quad b \in \mathcal{B} \quad (4)$$

to decide how many ambulances should be allocated to each base. Each term in the objective function, $\lambda_b \mathcal{E}(n_b, \lambda_b/\mu_b)$, corresponds to the expected number of lost calls offered to base b per unit time. As shown, for example, in Harel [15], $\mathcal{E}(n_b, \lambda_b/\mu_b)$ is a convex function of n_b , which implies that the objective function of Problem (1)–(4) is a separable convex function. Consequently, Problem (1)–(4) can be solved efficiently through simple marginal analysis; see Fox [9]. (One takes a greedy approach, adding ambulances one at a time, respecting the base capacities but otherwise choosing the base that leads to the greatest reduction in the objective function.) In certain situations, such as the ones considered in the next section, and in order to get a better qualitative understanding of this model, it will prove useful to relax the integrality requirement in (4) and use continuous version of the Erlang loss formula [17].

Recall that, in this model, calls that arrive when all ambulances at a base are busy are “lost.” The proportion of such calls is small when the load on a base is small. Hence, it is in this lightly loaded regime where the model is most appropriate.

The queueing model underlying the Erlang loss function assumes that the service times (including travel

time, time at scene, etc.) for successive calls are independent. This assumption does not hold precisely, because the locations of ambulances are affected by the locations of previous calls. The assumption is reasonable when a large portion of calls are served by ambulances that are already waiting at the base. As reported in Richards [25], even in systems having utilizations of around 35%, the fraction of calls responded to from the road is usually only around 10%. Furthermore, the time spent by an ambulance at the scene typically dominates the travel time [5], so the dependence between calls, even when calls are often responded to from the road, is mild.

The most unfortunate assumption in Problem (1)–(4) is that we have *a priori* knowledge of what portions of the demand are served by the different bases. In reality, the total rate of call arrivals into the system is known, but the proportions of the calls served by the different bases emerge as a result of the ambulance dispatch policy. Problem (1)–(4) assumes that fixed and known portions of the calls arriving into the system are offered to the different bases and the bases do not help each other even if they are not able to cope with their offered call loads.

3 Insights from the island model

Consider Problem (1)–(4) in the case where we have two bases, so that $\mathcal{B} = \{1, 2\}$. Our goal is to obtain insight into the optimal allocation of ambulances to bases. As we will see, allocating ambulances in proportion to the call loads at the bases can be highly suboptimal, but as the total load increases, this suboptimality decreases.

We first let $(\mu_1, \mu_2) = (1, 1)$ and $N = 20$, and solve Problem (1)–(4) for different values of (λ_1, λ_2) . We choose $(\mu_1, \mu_2) = (1, 1)$ only for convenience and other service rates can be captured by simply scaling our results. The regions and the associated labels in Fig. 1 show the optimal solutions to Problem (1)–(4) for different values of (λ_1, λ_2) . For example, if $(\lambda_1, \lambda_2) = (3, 2)$, then the optimal solution to Problem (1)–(4) is $(n_1, n_2) = (11, 9)$. In Fig. 1, the shaded area covers those regions for which allocating ambulances in proportion to call loads differs from the optimal allocation. It is easy to see that allocating ambulances in proportion to the call loads offered to the different bases is almost never optimal for low utilization regimes and it is only optimal for the higher utilization regimes if λ_1 is close to λ_2 . For example, if $(\lambda_1, \lambda_2) = (5, 2.5)$, then we have $\lambda_1/\lambda_2 = 2$, but the optimal solution to Problem (1)–(4) is $(n_1, n_2) = (12, 8)$ and $n_1/n_2 = 1.5$.

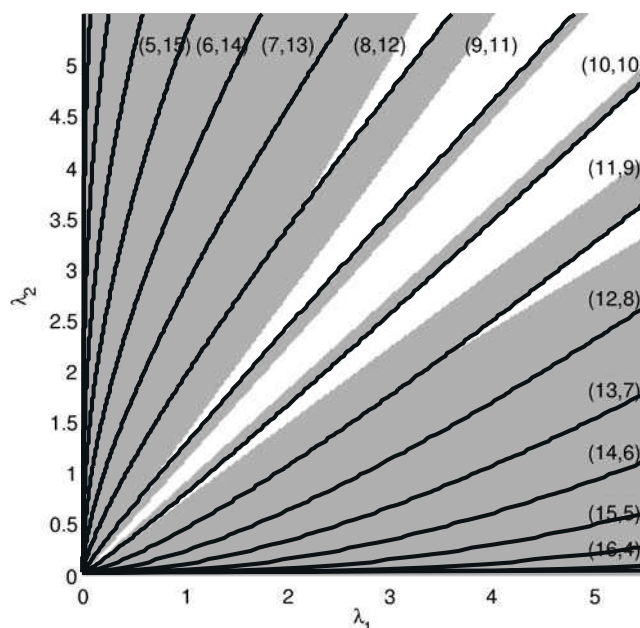


Fig. 1 Optimal solutions to Problem (1)–(4) for different values of (λ_1, λ_2)

Exploring further, suppose that $(\mu_1, \mu_2) = (1, 1)$ and $N = 10$. Since the service rates are equal to 1, the offered load per server is $\rho = (\lambda_1 + \lambda_2)/N$. We vary λ_1 but keep ρ and N fixed. In Fig. 2, we plot n_1^* , the optimal number of ambulances assigned to Base 1 as a function of λ_1 . We do this for 3 different values of ρ . In each case, the units of the x -axis have been linearly scaled and

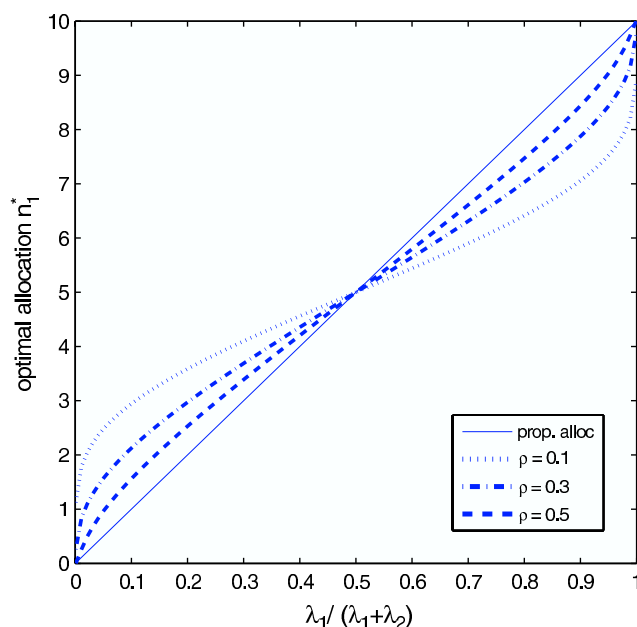


Fig. 2 Value of n_1 in the optimal solutions $(n_1, N - n_1)$ to Problem (1)–(4) for different values of $\lambda_1/(\lambda_1 + \lambda_2)$

correspond to $\lambda_1/(\lambda_1 + \lambda_2)$ which is the fraction of the total demand $\lambda_1 + \lambda_2$ that is offered to the first base. We note that Fig. 2 uses a continuous version of the Erlang loss formula to allow fractional ambulance allocations and this allows us to produce smooth plots [17].

If proportional allocation were optimal, we would have $n_1/N = \lambda_1/(\lambda_1 + \lambda_2)$ and the optimal solutions to Problem (1)–(4) as a function of $\lambda_1/(\lambda_1 + \lambda_2)$ would lie on the straight line in Fig. 2, for any value of the offered load per server. However, the figure shows that optimal solutions differ substantially from proportional allocation, especially when the offered load per server is low. When the offered load per server is high, proportional allocation is close to optimal. However, this requires offered load per server values as large as 0.5, whereas a more typical value in practice is 0.3; see, e.g., Budge et al. [5]. When $\lambda_1/(\lambda_1 + \lambda_2) < 0.5$, the number of ambulances allocated to the first base is larger than that suggested by proportional allocation. In other words, bases with lighter loads receive more than their proportional share of ambulances. But how significant is the sub-optimality of proportional allocation?

Figure 3 gives the loss rates for proportional allocation and the optimal allocation as a function of $\lambda_1/(\lambda_1 + \lambda_2)$. The results indicate that the cost of using proportional allocation can be very large. For example, if $\lambda_1/(\lambda_1 + \lambda_2) = 0.1$, then proportional allocation loses about 7% of the calls, whereas the optimal solution loses about 2% of the calls. This is certainly a practically significant difference. Similar to Fig. 2, Fig. 3 uses a continuous version of the Erlang loss formula.

The non-optimality of proportional allocation appears to be a general phenomenon, occurring in situations with more than 2 bases. Consider a situation with 6 ambulances and 3 bases with offered demands $(\lambda_1, \lambda_2, \lambda_3) = (1.2, 0.3, 0.3)$, respectively, so that $\rho = 0.3$. Proportional allocation yields $(n_1, n_2, n_3) = (4, 1, 1)$ which has a loss rate of 9.4% of calls, whereas $(n_1, n_2, n_3) = (3, 2, 2)$ has a loss rate of 7.1% of calls. Similar examples are easily constructed for greater numbers of ambulance and bases. To see why this observation should hold for more than 2 bases, consider the (unknown) optimal allocation of ambulances to bases. Select any 2 bases. The allocation of ambulances to these 2 bases must be optimal for a 2-base system, so our observation above should hold in this case, and the lower-loaded base receives a more-than proportional number of the ambulances shared by the 2 bases.

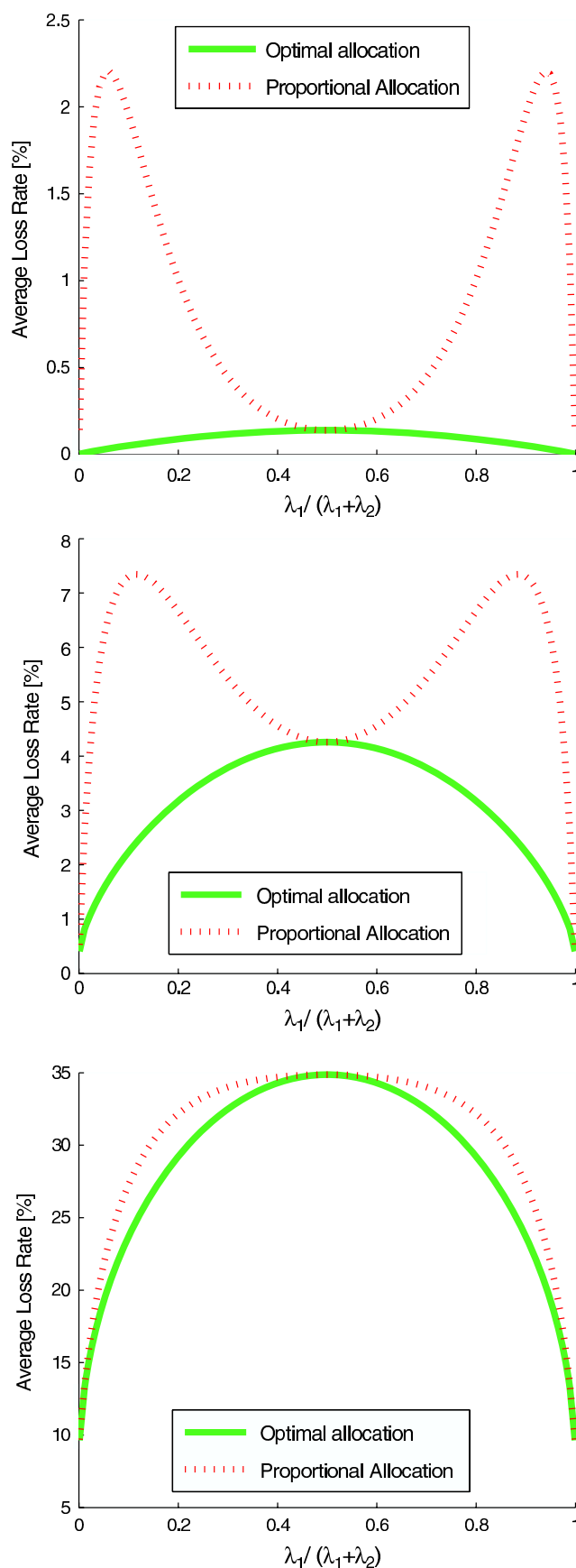


Fig. 3 Rate of lost calls for proportional allocation and the optimal allocation, for $\rho = 0.15, 0.3$ and 0.5 , respectively

4 The overflow model

In this section we propose a descriptive model (the overflow model) that estimates performance measures for a given ambulance allocation. The overflow model relaxes some of the critical assumptions of the island model in Section 2. In particular, it does not require that we know the call loads offered to different bases, and allows bases to cooperate.

The primary use we have in mind for the overflow model is to quickly evaluate a large number of possible ambulance allocations. We can then evaluate a small set of the most promising allocations through a detailed simulation study. Thus the overflow model is not meant to replace simulation but rather to complement it, by enabling pre-screening and fast discarding of many possible allocations.

We start by considering a simple extension of the island model meant to overcome the assumption that $\{\lambda_b : b \in B\}$ are known parameters. One can attempt to do this by turning these parameters into decision variables in Problem (1)–(4). This idea amounts to solving a problem of the form

$$\begin{aligned} \min \quad & \sum_{b \in B} \lambda_b \mathcal{E}(n_b, \lambda_b / \mu_b) \\ \text{subject to} \quad & (2), (3), (4) \\ & \sum_{b \in B} \lambda_b = \lambda \\ & \lambda_b \geq 0 \quad b \in B, \end{aligned}$$

where λ is the total rate of call arrivals into the system, and $\{n_b : b \in B\}$ and $\{\lambda_b : b \in B\}$ are the decision variables. This problem finds an allocation of the calls and ambulances among the bases to minimize the number of lost calls, and it mimics the natural load balancing sought after by dispatchers in practice. Unfortunately, numerical evaluations reveal that $\lambda_b \mathcal{E}(n_b, \lambda_b / \mu_b)$ is not jointly convex in (n_b, λ_b) . Therefore, there are no guarantees that the new problem has a unique local minimum, and finding a global optimum may be almost as difficult as enumerating all possible values of $\{n_b : b \in B\}$ that satisfy Constraints (2)–(4).

In view of this we consider a different approach that does not directly take optimizing the expected Erlang loss as the criterion for dividing the demands but instead views the problem as a queueing system with a spatial structure.

We start by assuming that there are a number of demand nodes indexed by $j = 1, \dots, J$. Each demand node j generates a Poisson arrival process of calls with rate d_j and the processes corresponding to different demand points are independent. The set of bases is $B = \{1, \dots, B\}$.

Each demand node j has a fixed priority ranking of all bases $j(1), \dots, j(B)$, which might correspond, for instance, to how far each base is from node j . A call originating at node j is first offered to an ambulance based at $j(1)$, and if no ambulance is available there, then it is offered to an ambulance based at $j(2)$, and so forth. If no ambulance is available at any base, then the call is lost and assumed to be answered by some alternative agent. The term “offered” emphasizes that some calls contribute to the load on a base, even if they are not actually served by that base. We say that a base is busy if all ambulances stationed there are busy.

Let s_{jk} denote the probability, under stationarity, that a call originating at node j will be answered by an ambulance from base $j(k)$. We extend the definition of s_{jk} to include $k = B + 1$, so that $s_{j, B+1}$ is the probability that no ambulance is available.

Here we point out that the probability that all ambulances are busy is typically extremely small. Indeed, if each ambulance is independently busy with probability ρ , where ρ is the overall utilization, then $s_{j, B+1} = \rho^N$. This value is negligible for systems with modest utilization and many ambulances. In general the true probability that all ambulances are busy is larger than this approximation, but still practically negligible, except in cases of extreme load that do not typically arise in daily operations.

Suppose now that we are able to obtain (approximations for) the probabilities s_{jk} . Then it is easy to compute (approximations for) any performance measure relating to the response time to calls, for which the conditional expected value given the originating demand point j and responding base k can be (approximated or) computed. Let ψ_{jk} denote this conditional expectation. Then the conditional expectation $\psi(j)$ given the originating demand point j is

$$\psi(j) := \sum_{k=1}^B s_{jk} \psi_{jk}. \quad (5)$$

Moreover, ψ is then given by

$$\psi = \frac{\sum_{j=1}^J d_j \psi(j)}{\sum_{j=1}^J d_j}. \quad (6)$$

Examples of such performance measures include the probability of reaching a call within a given time threshold, the expected response time and, as considered in Erdogan et al. [8], the survival probability.

In our analysis, we focus on the probability of reaching a call within a given time threshold, which we denote by r . Assuming deterministic travel times, if demand node j can be reached from, and only from, the

first $N(j)$ bases in demand node j 's list within the time standard, the specialization of Eq. (5) to this measure simplifies to

$$r(j) = \sum_{k=1}^{N(j)} s_{jk}. \quad (7)$$

In this setup, it is possible to derive a system of equations relating the probabilities s_{jk} and the node demands that has the form

$$S = f(S), \quad (8)$$

where S stands for the $J \times (B+1)$ matrix $[S]_{jk} = s_{jk}$ and f is a certain non-linear function.

We present the detailed derivation in Appendix A and only present the most important points here. First, the total demand offered to base b is the sum of a primary demand, coming from nodes for which b is the preferred base, plus an overflow demand coming from other nodes, when all bases preferred to b by those nodes are busy. The probability that all ambulances at a particular base are busy can be estimated from the demand offered to it using the Erlang loss formula. However, when trying to estimate the probability that a call from node j finds all servers at its k -th preferred base busy, conditioned on all ambulances in the previous $k-1$ bases to be busy, one has to take into account the addition to the demand offered to base k from overflow demand from the first $k-1$ bases. Careful estimation of the new demands resulting from this conditioning leads to a system of non-linear equations for the matrix of probabilities s_{jk} which can be put in the form (8).

5 Computational results for the overflow model, the A-Hypercube model and simulation

In this section, we present a case study based on data from the Edmonton, Alberta, EMS. The key performance measure considered is the percentage of calls wRTOTS, although we also look at vehicle utilization. We apply both the overflow model introduced in the previous section and the approximate hypercube (A-Hypercube) model developed in Larson [22]. We use the second method in Section 4 of Larson [22] to normalize the ambulance dispatch probabilities of the A-Hypercube model. Our goal is to verify that the performance measure estimates computed through the overflow model are accurate enough to reliably assist in identifying high-quality ambulance allocations for further exploration.

Some important structural differences between the A-Hypercube approximation scheme and the overflow model are worth mentioning here. In the A-Hypercube model, the fundamental descriptive variables are the individual vehicle utilizations $\rho = (\rho_a, a = 1, \dots, N)$, where N is the total number of ambulances. In the overflow model, the fundamental quantities are the probabilities s_{jk} that a call from any given zone is served by any given base. As in our model, the A-Hypercube estimation of ρ is based upon the solution of an equation of the form $\rho = f_H(\rho)$, where f_H is a certain non-linear function. In the original procedure, proposed by Larson [22], this is done by initializing $\rho^0 = (\bar{\rho}, \bar{\rho}, \dots, \bar{\rho})$, where $\bar{\rho} := \lambda/\mu N$ is the average global utilization of the system, and producing a sequence of estimates $\rho^1, \rho^2, \dots$, as follows. At each step, one first lets $\tilde{\rho}^k = f_H(\rho^{k-1})$ and then computes $\rho^k = \bar{\rho} \cdot (\tilde{\rho}^k) / (\sum_{a=1}^N \tilde{\rho}_a^k)$, thereby re-normalizing the utilizations to the average $\bar{\rho}$. The procedure we use to solve the fixed point Eq. (8) generates a sequence of matrices $S^0, S^1, \dots$ in an analogous way, but it does not enforce any normalization on S . The natural normalization condition, namely, that $\sum_k s_{jk} = 1$, for every j is automatically preserved by our f . See Appendix B for a discussion on ways to initialize the matrix S .

The A-Hypercube iterative procedure stops when $\|\rho^{k+1} - \rho^k\|_1 < \epsilon$, a given tolerance. In order to establish a fair comparison between the speeds of convergence for both methods, we set the stopping condition for the overflow model to be $\|R(S^{k+1}) - R(S^k)\|_1 < \epsilon$, where $R(S)$ is a function that computes individual vehicle utilizations from the probability matrix S . An exact description of this function can be found in Appendix C.

Our input data are taken from Budge et al. [5], describing the ambulance system operating in Edmonton, Alberta. The system is medium sized with 16 bases and 180 demand nodes. We experimented with allocations of 12 and 16 ambulances and demand rates ranging between 2 and 7 calls per hour. The service rate is about $\mu = 0.97$ calls per hour per ambulance, which corresponds to an average service time of about 62 minutes per call.

A computation shows that there are approximately 400,000 different ways of allocating 12 ambulances among 16 bases given the capacity constraints for the bases in this system. The number of allocations of 16 ambulances is also very close to 400,000. In our experimental setup, we select subsets of 1,000 allocations among the 400,000 possible ones, for each of these two scenarios. We apply both the overflow model and the A-Hypercube model to each of these allocations to estimate the base (resp. vehicle) utilizations. From these

we calculate predictions for the system-wide percentage of calls wRTOTS as

$$100\% - \frac{\sum_{j=1}^J d_j r(j)}{\sum_{j=1}^J d_j},$$

where $r(j)$ is given by (7), and d_j is the demand rate at location j .

To provide a benchmark for comparison, we use discrete-event simulation to compute the same performance measure for each of the 1,000 ambulance allocations in a simplified model of ambulance operations. The specifics of the simulation model are as follows. Calls are generated in each of the 180 locations as independent Poisson processes with constant rates equal to the historical demand rates. All calls are assumed urgent. The ambulance dispatching policy is to send the closest available ambulance to a call, and if there are no available ambulances, then the call is put in a first-in-first-out queue. (Putting a call in a queue almost always implies that the call will not meet the time standard.) In the simulation, travel times between each base and demand zone are deterministic and taken to be equal to the average travel time observed from historical data. Ambulances spend an exponentially distributed amount of time at the scene with a mean of 12 minutes. Patients require transport to a hospital with probability 0.75. Given that a patient is to be transported to a hospital, the choice of hospital is independent of all else and chosen according to historical probabilities. The time spent at the hospital is Weibull distributed with a mean of 30.4 minutes and a shape parameter of 2.5.

For each allocation, we simulate the system for two weeks and make 50 runs to estimate the fraction of calls wRTOTS. Fifty replications are enough to estimate this fraction with a 95% confidence interval of width 0.6%. More importantly, since we are using common random numbers (i.e., the same set of calls) for all allocations, 50 replications is enough to establish a difference between allocations whose mean performances differ by 0.5% or more. For instance, in the experiment with 12 ambulances and $\lambda = 3$ calls per hour, the simulation of allocations #345 and #435 yields mean costs of $16.7\% \pm 0.3\%$ and $16.5\% \pm 0.3\%$, respectively, and a difference of $0.3\% \pm 0.3\%$. Thus, the second allocation is better than the first one with high confidence, even when the confidence intervals for their mean costs have a big overlap. The average response time is, of course, a function of the allocation. For the best allocation in the situation above, the average response time is 5.8 min. The average time of transport to the hospital is observed to be 16.4 min, which is considerably larger due to the fact that there are significantly fewer hospitals

than bases and that the ambulance travels at regular speed when driving to the hospital.

The chart on the left-hand side of Fig. 4 shows a scatter plot of the estimate of the fraction of calls wRTOTS as predicted by the overflow model versus the corresponding fraction measured from the simulation model for 1,000 ambulance allocations. The total number of ambulances is 12, and the total call rate is $\lambda = 3$ calls/hr which corresponds to an average utilization $\bar{\rho} = 0.26$. The chart on the right-hand side of the same figure shows the scatter plot of estimates yielded by the A-Hypercube model versus those from the simulation model, for the same 1,000 allocations. The solid line on both charts is the line $y = x$ on which the points would

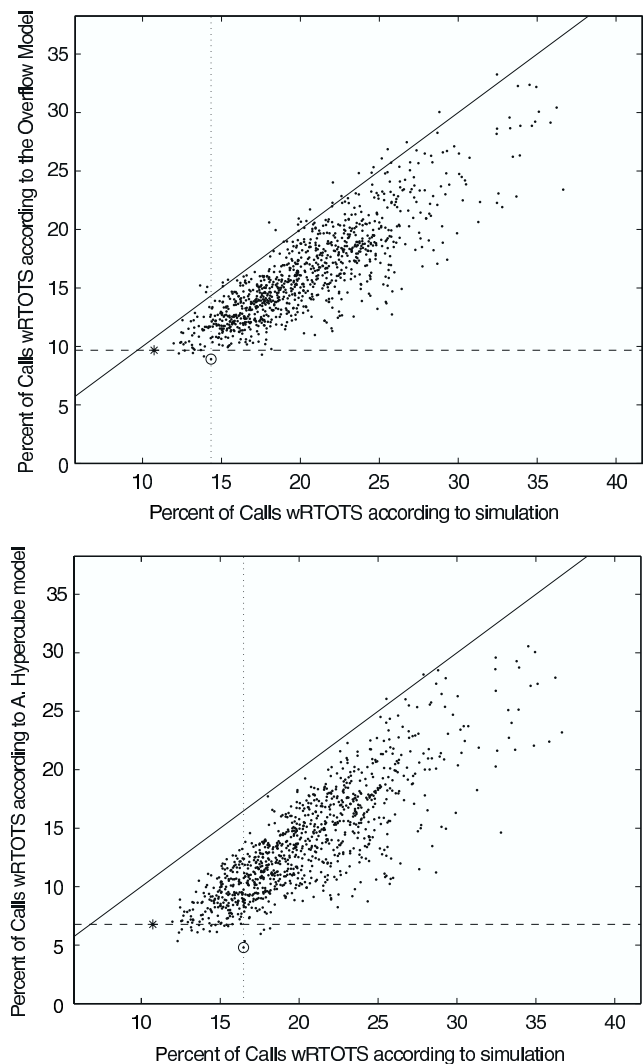


Fig. 4 Scatter plot of the fraction of calls wRTOTS estimated by the overflow model – left (resp. the A-Hypercube model – right) versus the fraction determined through simulation. The figures are percentages of the total rate of calls

Table 1 Comparison of accuracy measures of the overflow and A-Hypercube models for a range of average utilizations

Sim. params.			Overflow model				A-hypercube model				
<i>N</i>	λ	$\bar{\rho}$	R. Corr	RMSE	<i>F</i>	<i>G</i>	R. Corr	RMSE	<i>F</i>	<i>G</i>	O.D.
12	2	0.17	0.85	*5.8%	*0.006	0.004	0.83	6.6%	0.025	0.004	0
16	3	0.19	*0.86	*4.2%	0.001	*0.003	0.82	5.1%	0.002	0.018	1
12	3	0.26	0.84	*4.8%	*0.006	*0.046	0.82	7.2%	0.020	0.190	0
16	5	0.32	0.73	*2.5%	0.090	0.261	0.74	6.7%	*0.020	0.301	3
12	5	0.43	0.71	*4.0%	0.122	0.060	0.79	7.8%	*0.014	0.052	2
16	7	0.45	0.49	*6.5%	0.457	*0.035	*0.67	7.5%	*0.033	0.191	0

An * indicates an instance where a measure for a model is significantly better than the same measure for the other model

lie if the predictions of the approximate models agreed perfectly with the simulation measurements.

The overflow model predicts the fraction of calls wRTOTS more accurately than the A-Hypercube model. The predictions of both models exhibit a negative bias, with the A-Hypercube model bias being more pronounced. The root mean-squared error (RMSE) of the overflow model predictions is 4.8% whereas the RMSE of the A-Hypercube model predictions was 7.2%. The Spearman's (rank) correlation between the overflow model's predictions and the simulation measurements is 0.86, which is slightly better than that for the A-Hypercube model (0.82).

As stated earlier, the primary purpose of the overflow model is to serve as a pre-screening tool to evaluate a large set of possible allocations quickly in order to identify the most promising ones to evaluate using detailed simulation. To investigate the effectiveness of this approach we define *F* as the fraction of allocations that have a model-predicted cost below that of the allocation having minimum simulated cost. The usefulness of measure *F* is that it is proportional to the number of allocations one has to evaluate through simulation in increasing order of model-evaluated cost before one evaluates the optimal allocation. In the charts in Fig. 4 the allocation having minimum simulated cost is marked with a * and the fraction just defined corresponds to the fraction of allocations below the horizontal dashed line. For the overflow model we have $F = 0.006$. For the A-Hypercube model we have $F = 0.020$, which is considerably larger.

The results we have just described are not uncommon in the low to medium-low utilization range. Table 1 summarizes the results of a series of experiments carried out on allocations having 12 and 16 ambulances over a range of system utilizations. In general we see that the overflow model has greater accuracy in predicting the percentage of calls wRTOTS, as evidenced by lower value of the RMSEs throughout. Rank correlations (R. Corr.) are similar for both models, with the A-Hypercube being significantly better only for the highest utilizations. The overflow model *F*-fractions are

comparatively lower than those of the A-Hypercube for the lowest utilizations but higher for the highest utilizations.

Another measure registered in Table 1, of somewhat lesser importance than *F*, is *G*, defined as the fraction of allocations having simulated cost lower than that of the best model-ranked policy. In the charts, this corresponds to the fraction of points to the left of the dotted line. This fraction is a measure of how far the best model-ranked policy is from optimality. A notion of the magnitude of this quantity could be helpful for ambulance managers who do not have the means to evaluate allocations through detailed simulation but only have a model upon which to rely. The overflow model yields *G*-ratios that are at least as low as those yielded by the A-Hypercube model in 5 of the six cases presented in Table 1 and yields significantly lower values in 3 of them.

Another important measure of the quality of these methods is how well they predict the utilization of individual vehicles. The chart on the left-hand side of Fig. 5 shows a histogram of the absolute difference between the individual vehicle utilizations predicted by the overflow model and those obtained by simulation. The chart on the right does the same for the A-Hypercube model predicted utilizations. The differences are taken individually for each of the vehicles and for each of the 1,000 evaluated allocations. A positive difference indicates that the model-predicted utilization is larger than the simulated one. The utilization differences for the overflow model have a mean of 2.5% and a standard deviation of 2.9% whereas those of the A-Hypercube model are less biased, with a mean of 0.6%, but less precise, with a standard deviation of 4.8%. The situation exemplified here is not atypical. As Table 2 shows, for the same set of scenarios as before, overflow-model predicted utilizations have a positive systematic error that grows as the system utilization grows but smaller standard deviations than those yielded by A-Hypercube model predictions. On the other hand, the A-Hypercube predictions have a very small bias. This is not surprising considering

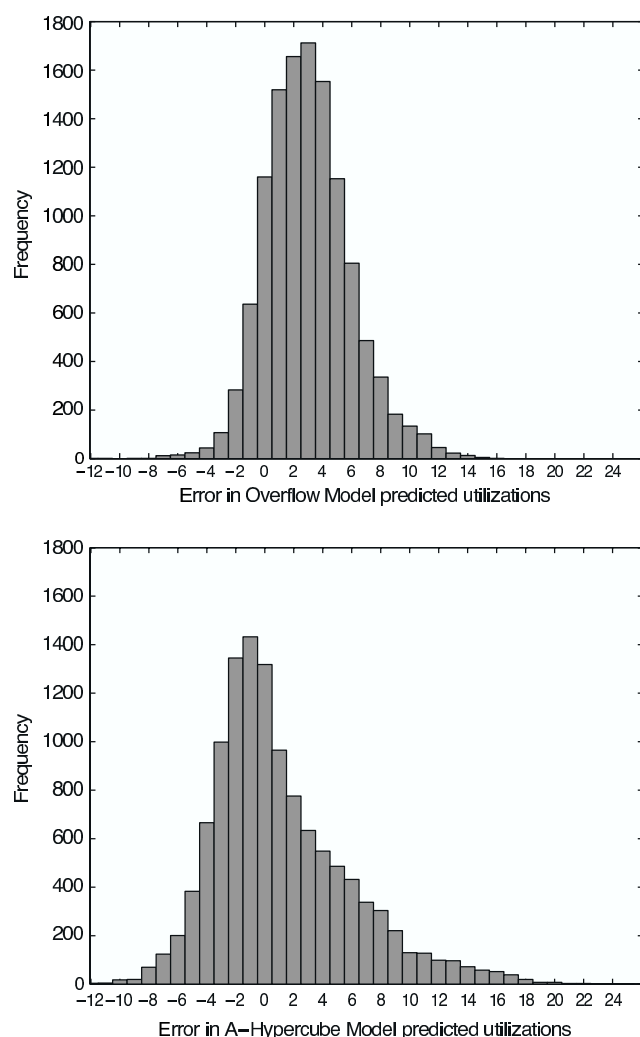


Fig. 5 Differences between vehicle utilizations obtained from the overflow model and the simulation model (left). Corresponding differences for the A-Hypercube model

that, as mentioned earlier, the A-Hypercube procedure renormalizes base utilizations to the system utilization at each step.

We briefly comment now on the computational speed of both methods. The fixed point procedures were implemented in MATLAB and run on a 3.2 GHz Pentium IV PC running Linux. For the situation with 12

ambulances and $\lambda = 3$ calls/hr, each iteration of the A-Hypercube solution method takes 7.8 ± 2.5 ms whereas each iteration of the overflow method took 240 ± 65 ms. Thus, in this implementation each iteration of the A-Hypercube is faster by a factor of 30. This is expected considering that each iteration of the overflow method updates a whole matrix of probabilities of size $J \times B$, whereas the A-Hypercube method deals with a vector of size N , which is generally much smaller. We also note that the MATLAB implementation of the A-Hypercube method was easy to vectorize while this was not possible for the overflow method. We anticipate that the speed ratio of C language implementations of the methods would be closer to 10:1 or even less. The average number of iterations taken by each of the methods to converge was, for $\epsilon = 10^{-6}$, 11.7 ± 2.1 (A-Hypercube) and 16.9 ± 1.7 (overflow). For $\epsilon = 10^{-4}$, we got means of 6.7 and 10.9 iterations, respectively. The A-Hypercube method failed to converge in 2 out of the 1000 allocations studied while the overflow method converged for all of them.

Finally, we note that both ways of initializing the matrix S^0 discussed in Appendix B led to the same fixed point in all cases. We conjecture that the function f implicitly defined by (9)–(14) has a unique fixed point, but this is still an open question.

6 Conclusions and future research

We introduced two models for the static ambulance deployment problem. The models capture some of the essential queueing dynamics of an emergency medical service system while allowing efficient solution procedures. The island model is particularly suitable when analyzing multi-region systems managed by a central planner. It illustrates that ambulances should not be allocated in proportion to the loads offered to the locations, especially when ambulance utilizations are low. It is primarily intended to provide insight into ambulance-allocation decisions, and its recommendations should be viewed as a starting point for further exploration of allocations.

The overflow model estimates the loads offered to the different bases through a system of equations involving the Erlang loss formula. Whether this system of equations has a unique solution or not is an open question. Computational experiments suggest that the predictions of the second model are quite accurate, and the model is particularly useful for evaluating a large set of possible ambulance allocations and selecting some allocations to be further evaluated through simulation. The method is computationally more demanding than

Table 2 Error statistics for individual vehicle utilizations according to the two models

Sim. params		Overflow model		A-hypercube model	
N	λ	Error	Std.	Error	Std.
16	3	1.3%	2.5%	0.7 %	4.0%
12	3	2.5%	2.9%	0.6 %	4.8%
16	5	5.4%	3.6%	1.0 %	5.6%
12	5	9.6%	3.9%	0.2 %	5.8%
16	7	13.3%	4.5%	0.9 %	6.6%

the A-Hypercube model, but still orders-of-magnitude faster than simulation. Its predictions are more accurate than those of the A-Hypercube for light to moderate ambulance utilization, while the situation is reversed for moderate to heavy ambulance utilization.

It would be interesting to investigate whether our models can incorporate the fact that the average service time for an ambulance stationed at a base is affected by the location of the demand assigned to it. Also, provided the necessary input data is available, it would be straightforward to take random travel times into account in our analysis and test whether this feature improves the accuracy of the predictions. In particular, if the distribution of travel times from any base k to any node j is known, one can compute any steady state performance measure that is a function of the travel time to the call by conditioning on j and k in Eq. 5.

An important assumption underlying both of our models is that demand is independent at different locations. This seems reasonable when the ambulance service is operating under typical conditions. But the assumption is violated when one considers the possibility of catastrophic events that strike multiple locations simultaneously. To effectively plan for such situations, it would be necessary to consider fundamentally different strategies.

In summary, we have developed analytical tools to aid EMS system managers in their quantitative evaluations of ambulance allocation decisions. We believe that our tools will be most valuable in pointing out the promising ambulance allocations. In this sense, we do not expect our tools to completely replace simulation, but rather to complement it, by enabling pre-screening and fast discarding of many poor ambulance allocations.

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Appendix A: Detailed derivation of approximation procedure

In this appendix we provide the derivation of a system of nonlinear equations that will yield, upon its solution, the probabilities s_{jk} that a call produced at zone j will be served by an ambulance stationed at base $j(k)$.

Define, for all $k = 1, \dots, B + 1$, $\mathcal{A}_{jk}(t)$ to be the event that the first $k - 1$ bases in demand node j 's ranking of bases are busy at time t . In other words, $\mathcal{A}_{jk}(t)$ is the event that none of the bases preferred to the k -th base in demand node j 's list has an available ambulance at time t . The event $\mathcal{A}_{j1}(t)$ has probability 1 by definition. If a call is generated at demand node j at time t , then on the event $\mathcal{A}_{jk}(t)$, the call is offered to base $j(k)$. We write $\mathcal{A}_{jk}$ for $\mathcal{A}_{jk}(0)$ and let $a_{jk} = P(\mathcal{A}_{jk})$ be the stationary probability of the event $\mathcal{A}_{jk}(t)$. We can interpret a_{jk} as the probability that a call originating at demand node j will be offered to base $j(k)$. (We are implicitly using a Poisson-arrivals-see-time-averages argument here to ensure that the Poisson arrivals at demand node j see the time-average behavior of the system. We also use the fact that the time-average behavior corresponds with the stationary behavior. For background on both of these topics, we refer the reader to Wolff [29].) Similarly, we let $\mathcal{S}_{jk}(t)$ denote the event that the first available ambulance in the priority list of bases $j(1), \dots, j(B)$ at time t is at $j(k)$, and write $\mathcal{S}_{jk}$ for $\mathcal{S}_{jk}(0)$. Thus, as defined in Section 4 $s_{jk} = P(\mathcal{S}_{jk})$ is the probability that a call originating at node j will be answered by an ambulance from base $j(k)$. We extend the definition of s_{jk} to include $k = B + 1$, so that $s_{j,B+1}$ is the probability that no ambulance is available anywhere.

We therefore turn to developing an approximation for the probabilities s_{jk} for all $j = 1, \dots, J$, $k = 1, \dots, B$. An approximation for $s_{j,B+1}$ will follow from the fact that $\sum_{k=1}^{B+1} s_{jk} = 1$, this being the probability that a call is either answered by one of the bases or lost.

We begin by noting that $\mathcal{A}_{j,k+1} \subseteq \mathcal{A}_{jk}$ and $\mathcal{S}_{jk} = \mathcal{A}_{jk} \setminus \mathcal{A}_{j,k+1}$ for all $k = 1, \dots, B$. In other words, the first k bases on demand node j 's list can be busy only when the first $k - 1$ bases are busy, and a call is answered by the k -th base in demand node j 's list if and only if the first $k - 1$ bases are busy, but the k -th base is not. Hence, for $k = 1, \dots, B$, $s_{jk} = a_{jk} - a_{j,k+1}$, $\mathcal{A}_{jk} = \bigcup_{i=k}^{B+1} \mathcal{S}_{ji}$, and $a_{jk} = \sum_{i=k}^{B+1} s_{ji}$ for all $k = 1, \dots, B$. We let A and S respectively be the matrix of values $\{a_{jk} : j = 1, \dots, J, k = 1, \dots, B + 1\}$ and $\{s_{jk} : j = 1, \dots, J, k = 1, \dots, B + 1\}$. Since we can recover either of these matrices from the other one, we focus on computing A through a fixed point equation.

Recall that $a_{j1} = 1$ for all $j = 1, \dots, J$. To approximate a_{j2} , we first let λ_b be the “offered” load to base b , so that

$$\lambda_b = \sum_{j=1}^J d_j a_{j,j^{-1}(b)}, \quad (9)$$

where $j^{-1}(b)$ stands for base b 's index in node j 's ranking. We then approximate a_{j2} , the probability that no ambulances are available at base $j(1)$, by

$$a_{j2} \approx \mathcal{E}(n_{j(1)}, \lambda_{j(1)}/\mu_{j(1)}). \quad (10)$$

(The number of ambulances n_k and the service rate μ_k at base k are constants that do not change in our scheme.) It now remains to show how to compute approximations for a_{jm} for $m > 2$. The nested nature of the events $\mathcal{A}_{jm}$ for all $m = 1, \dots, B$ ensures that we have

$$a_{jm} = P(\mathcal{A}_{j2}) P(\mathcal{A}_{j3}|\mathcal{A}_{j2}) \cdots P(\mathcal{A}_{jm}|\mathcal{A}_{j,m-1}) \quad (11)$$

for $m > 2$. We approximate each factor of the form $P(\mathcal{A}_{j,k+1}|\mathcal{A}_{jk})$ by

$$P(\mathcal{A}_{j,k+1}|\mathcal{A}_{jk}) = \mathcal{E}(n_{j(k)}, \lambda(j, k)/\mu_{j(k)}) \quad (12)$$

for all $k = 2, \dots, B$, where $\lambda(j, k)$ represents the conditional demand rate offered to base $j(k)$, given $\mathcal{A}_{jk}$. By conditioning on $\mathcal{A}_{jk}$, the demand offered to base $j(k)$ may not be Poisson, but we ignore this issue in our approximation. In analogy with (9), the conditional demand can be written as

$$\lambda(j, k) = \sum_{i=1}^J d_i P(\mathcal{A}_{i, i^{-1}(j(k))}|\mathcal{A}_{jk}), \quad (13)$$

where $i^{-1}(j(k))$ is defined as the n for which $i(n) = j(k)$, i.e., the index of base $j(k)$ in demand node i 's list of bases. The probability in (13) is the conditional probability that a call originating at demand node i will be offered to base $j(k)$, conditional on the first $k-1$ bases in demand node j 's priority list being busy.

If all of the bases that appear before base $j(k)$ in demand node i 's list also appear in demand node j 's list before base $j(k)$, then we know that they are all busy, since we are conditioning on $\mathcal{A}_{jk}$. In that case, the conditional probability that the call originating at demand node i will be offered to base $j(k)$ is 1. If not, then we essentially need to compute the probability that a subset of the bases are busy, given that another subset of the bases are busy. In principle, it is possible to include such probabilities as separate variables and to write linking equations for them. However, this would cause a dramatic increase in the number of variables. To produce a tractable set of equations, we instead adopt the following approximation.

For given demand nodes i, j and a given base $j(k)$, we let $n := i^{-1}(j(k))$, i.e., the index of base $j(k)$

in node i 's list. In this case, since $\mathcal{A}_{in} = \cup_{m=n}^{B+1} S_{im}$, we have

$$\begin{aligned} P(\mathcal{A}_{in}|\mathcal{A}_{jk}) &= \frac{P(\mathcal{A}_{in} \cap \mathcal{A}_{jk})}{P(\mathcal{A}_{jk})} \\ &= \frac{1}{P(\mathcal{A}_{jk})} P(\cup_{m=n}^{B+1} S_{im} \cap \mathcal{A}_{jk}). \end{aligned}$$

On the event $\mathcal{A}_{jk}$, bases $j(1), \dots, j(k-1)$ are busy. Therefore, if base $i(m)$ is contained in this list, then base $i(m)$ cannot answer the call originating at demand node i . In such a case, we have $S_{im} \cap \mathcal{A}_{jk} = \emptyset$. Consequently, we define the set $M = \{m : n \leq m \leq B+1, i(m) \notin \{j(1), \dots, j(k-1)\}\}$ of bases that are not necessarily busy on the event $\mathcal{A}_{jk}$. In this case, we have

$$P(\mathcal{A}_{in}|\mathcal{A}_{jk}) = \frac{1}{P(\mathcal{A}_{jk})} P(\cup_{m \in M} S_{im} \cap \mathcal{A}_{jk}), \quad (14)$$

where the numerator essentially states that a call originating at i must be answered by a base that is not busy, and therefore, not in the list of busy ambulances $j(1), \dots, j(k-1)$. Finally, letting $c \wedge d = \min\{c, d\}$, we make the crude approximation that

$$\begin{aligned} P(\cup_{m \in M} S_{im} \cap \mathcal{A}_{jk}) &\approx (\sum_{m \in M} P(S_{im})) \wedge P(\mathcal{A}_{jk}) \\ &= (\sum_{m \in M} S_{im}) \wedge a_{jk}, \end{aligned}$$

which amounts to neglecting the “higher order” correction to $P(S_{im})$ that comes from intersecting with $\mathcal{A}_{jk}$. Taking the minimum with a_{jk} ensures that (14) yields a result in $[0, 1]$.

Combining (9)-(14) with the (linear) equations relating s_{jk} to a_{jk} , we obtain a system of equations for the entries of the matrix S that has the form $S = f(S)$ for an appropriately specified function f . Thus, S is a fixed point of the system, and once we determine S , we can compute performance measures as described earlier.

Appendix B: Initializing the iterative procedure for solving the overflow model fixed-point equations

Here we comment on different ways to initialize the matrix S^0 at the start of the iterative procedure described in Section 4.

The solution matrix should satisfy $\sum_{k=1}^{B+1} s_{jk} = 1$ for every $j = 1, \dots, J$, i.e., it is stochastic. So it is natural to choose an S^0 having this property as well. We tried 2 possibilities.

1. For each $j = 1, \dots, J$ let $s_{j1}^0 = 1$ and $s_{jk}^0 = 0$ for $k = 2, \dots, B+1$, i.e., allocate all of the demand coming from j to its preferred base.

2. For each $j = 1, \dots, J$ let $s_{jk}^0 = \bar{\rho}^{\sum_{l=1}^{k-1} n_{jl}b} (1 - \bar{\rho}^{n_{jk}b})$, for $k = 1, \dots, B$, and $s_{j,B+1}^0 = 1 - \sum_{k=1}^B s_{jk}^0$ where $\bar{\rho} := \lambda/N\mu$ is the average system utilization. Under the independent server approximation, the value of s_{jk}^0 corresponds to the probability that all bases preferred to $j(k)$ in node j 's ranking are busy but base $j(k)$ is not.

Appendix C: Detailed definition of function $R(\cdot)$

Let $R(S) := (R(S)_1, R(S)_2, \dots, R(S)_N)$ where $R(S)_v$, for $v = 1, \dots, N$ represents the utilization of ambulance v . Denote by $b = b(v)$ the base at which ambulance v is stationed, and by $\lambda_b = \lambda_b(S)$, the total demand offered to this base, as defined by Eq. 9; recall from Appendix A that $a_{jk} = \sum_{i=k}^{B+1} s_{ji}$. Then, according to the Erlang loss model, the utilization of each of the n_b ambulances stationed at base b equals the average expected utilization

$$R(S)_v = \frac{1}{n_b} \sum_{k=1}^{n_b} k P_{\mathcal{E}}(n_b, \lambda_b/\mu, k),$$

where

$$P_{\mathcal{E}}(n, \rho, k) = \frac{\rho^k/k!}{\sum_{j=0}^n \rho^j/j!}$$

is the Erlang probability that k out of the n ambulances are busy.

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